

Arav Raghunathan

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Electrical Engineering, Minor in Computing & Intelligence

Atlanta, GA

Expected in May 2028

EXPERIENCE

Software Engineering Intern

SoloPulse Co.

May 2026 - August 2026

Peachtree Corners, GA

- Independently architected and developed the complete proof-of-concept system for a U.S. Army ADAS contract as the sole engineer, driving system-level design decisions and collaborating with engineering leadership through iterative technical reviews.
- Engineered a distributed **gRPC-based network** connecting a NUC to 6 NVIDIA Jetsons, building a C++ system manager and API controller with a custom TCP image streamer to support real-time status monitoring and image streaming.
- Configured dnsmasq and systemctl across **6+** Linux edge devices, improving deployment workflows and supporting scalable distributed-system operation within a collaborative engineering environment.

Undergraduate Research Assistant

ICE Lab @ Georgia Tech

January 2026 – Present

Atlanta, GA

- Designing and developing **Field-Programmable Analog Arrays (FPAAs)** in SkyWater's 130 nm CMOS process for Vector Matrix Multiplication (VMM), supporting the lab's goal to develop efficient ML systems.
- Simulating CMOS circuits in Xschem and Ngspice and creating corresponding layouts in Magic, translating designs from schematic development through physical implementation and layout verification.
- Verifying design integrity using **LVS**, ensuring alignment between schematic and fabricated layout.

Peer Instructor

The Hive Makerspace

January 2026 – Present

Atlanta, GA

- Supported a high-traffic makerspace serving **1,000+** students, advising on hands-on technical skills including breadboarding and PCB fabrication.
- Mentored students on 3D printing (Bambu, Resin), circuit analysis, and laser cutting, fostering hands-on creativity and technical confidence.

ASIC Physical Design Engineer

SiliconJackets

January 2026 - Present

Atlanta, GA

- Designing and implementing a **32-bit RTL calculator** in SystemVerilog, developing a custom datapath architecture and SRAM-based result storage while applying digital design, computer architecture, and RTL development principles.
- Supporting **ASIC tapeout workflows** as part of the Physical Design team, collaborating on processes to meet project deadlines.
- Developing advanced proficiency in **SystemVerilog**, Cadence Virtuoso, and Innovus Stylus, applying these tools to complex digital design and physical implementation tasks.

Software Development Intern

Oracle Corporation

August 2024 - May 2025

Suwanee, GA

- Executed collaborative software projects utilizing **Oracle APEX**, developing resources for financial tracking and organization.
- Researched and synthesized industry papers on cloud computing's impact on healthcare, presenting to build shared understanding of emerging industry trends.

PROJECTS

32-Bit RTL Calculator | SystemVerilog, Cadence Design, Innovus Stylus, Python, Linux

- Engineered the conceptual and physical design of a calculator that performs bit-wise addition.
- Leveraged SystemVerilog and Innovus Stylus to develop RTL functionality and implement testbench-based verification methodologies to validate design correctness.

TECHNICAL SKILLS

Programming Languages: Java, Python, MATLAB, C, C++, RISC-V, SQL, PL-SQL, SystemVerilog, Linux, RTL, VHDL

Hardware Design & Embedded Systems: Xschem, Ngspice, Magic, LVS (Layout Versus Schematic), Breadboarding, Waveform Analysis, Arduino, ESP32, FPGA/FPAA Design, Signal Processing, Cadence, Synopsys, Innovus Stylus

Tools & Softwares: PTC Onshape, Oracle Application Express, Microsoft Office, Git, GDB, GCC, Dnsmasq, Systemctl